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Supporting Information for

Physics-tailored machine learning reveals unexpected physics in dusty plasmas

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Supporting Information Text

Details of the model structure. Our model is implemented in TensorFlow (1). In this section, all the bold text are functions in TensorFlow, with input parameters in the bracket after the function, if necessary. If parameter values are not mentioned, they are assumed to be the default. As described in the main text, the model consists of 3 neural networks (NNs) trained in parallel: g_{int}, $\vec{g}_{\text{env}}$, and g_γ. Both g_{int} and $\vec{g}_{\text{env}}$ have 3 dense-connected hidden layers, with **he_normal** initialization and **L2** regularization. The network g_{int} has 32 neurons for each hidden layer with **leakyrelu** ($\alpha = 0.1$), **tanh**, and **leakyrelu** ($\alpha = 0.1$) as activation functions, respectively. The last hidden layer is fully connected to a single output, the magnitude of the reduced interaction force in the xy plane, multiplied by horizontal separation, $f_{ij}\rho_{ij}$. The multiplication of the force by ρ_{ij} serves two purposes. The first is to lessen the divergence of the output as $\rho_{ij} \to 0$. The second is to save considerable computing time by not calculating a square root for every interaction force vector, which would be calculated for each particle interaction pair:

$$\vec{f}_{ij} = f_{ij}\hat{\rho}_{ij} = \frac{f_{ij}\rho_{ij}\vec{\rho}_{ij}}{\rho_{ij}^2}. \quad [S1]$$

The network $\vec{g}_{\text{env}}$ has 16 neurons for each hidden layer with **elu**, **tanh**, and **elu** as activation functions, respectively. The last hidden layer is fully connected to two outputs, $f_{i,x}^{\text{env}}$ and $f_{i,y}^{\text{env}}$. Finally, the network g_γ has 2 hidden layers with 16 neurons each, and **elu** and **tanh** as activation functions, respectively. The last hidden layer is fully connected to a single output: the damping coefficient, γ_i. As described in the main text, our model fits the reduced net force, $\sum_j \vec{f}_{ij} + \vec{f}_i^{\text{env}} - \gamma_i\dot{\vec{\rho}}_i$, to each particle's experimental acceleration, $\ddot{\vec{\rho}}_i$.

To reduce the amplification of measurement error by temporal differentiation, we apply the weak form (2) in our loss function:

$$L = \frac{1}{2N_pT_{\text{train}}} \sum_{i=0}^{(N_p-1)} \sum_{t\in\mathbb{T}_{\text{train}}} \sum_{\alpha}^{\{x,y\}} L_{i,t,\alpha}, \quad [S2]$$

$$L_{i,t,\alpha} = H\left(w \circledast_t (\vec{f}_i^{\text{env}} + \sum_j \vec{f}_{ij} - \gamma_i\dot{\vec{\rho}}_i - \ddot{\vec{\rho}}_i)_\alpha; \delta\right) \quad [S3]$$

Here T_{train} is the total number of frames for the particle trajectories in the training dataset, $\mathbb{T}_{\text{train}}$, N_p is the number of particles, $\alpha \in \{x, y\}$, and w is a customized weight function, defined in the range $[-\tau\Delta/2, \tau\Delta/2]$:

$$w_{t'} = w(t'\Delta) = \frac{30}{(\tau\Delta)^5}\left((t'\Delta)^2 - (\tau\Delta/2)^2\right)^2, \quad [S4]$$

where the recording time step $\Delta = 0.005$ s, and $\tau = 16$ is the size of the convolution window. The function H is a Huber loss function that reduces the relative weight of outliers in the loss function. The parameter δ controls the threshold of this reduction. The convolution function $\circledast_t$ is defined as:

$$a \circledast_t b = \Delta\int_{-\tau/2}^{\tau/2} a(t'\Delta)b(t'\Delta + t\Delta)\,dt' = \Delta\sum_{t'=-\tau}^{\tau} S_{t'}a_{t'}b_{t+t'}. \quad [S5]$$

In the last step of the equation above, Simpson discretization is used to compute the integral over each window, with the coefficient:

$$S_{t'} = \begin{cases} 1/3, & \text{if } |t'| = \tau \\ 4/3, & \text{if } |t'| < \tau \text{ and } (t'+\tau) \text{ is odd} \\ 2/3, & \text{if } |t'| < \tau \text{ and } (t'+\tau) \text{ is even} \\ 0, & \text{else.} \end{cases} \quad [S6]$$

By definition, at $t' = \pm\tau/2$, $w(t') = 0$ and $\dot{w}(t') = 0$. Therefore, it is easily proven through integration by parts that:

$$w \circledast_t \dot{\vec{\rho}}_i = -\dot{w} \circledast_t \vec{\rho}_i \quad [S7]$$

$$w \circledast_t \ddot{\vec{\rho}}_i = \ddot{w} \circledast_t \vec{\rho}_i \quad [S8]$$

As a result, our loss function becomes:

$$L_{i,t,\alpha} = H\left((w \circledast_t \vec{f}_i^{\text{env}} + \sum_j w \circledast_t \vec{f}_{ij} + \gamma_i\dot{w} \circledast_t \vec{\rho}_i - \ddot{w} \circledast_t \vec{\rho}_i)_\alpha; \delta\right) \quad [S9]$$

Thus, by using the weak form, temporal derivatives of experimental particle positions are replaced by derivatives of the weight function, which is analytic.

As mentioned previously, the parameter δ controls the crossover from quadratic to linear loss in the **Huber** loss function, H, which is implemented in TensorFlow. For a fitting residue e_i, when $e_i < \delta$, $H(e_i;\delta) = e_i^2/2$ and when $e_i > \delta$, $H(e_i;\delta) = \delta e_i - \delta^2/2$.

Considering that a very large fitting error on a single data point might arise from other sources of noise (for example, tracking error), this large error should be deemphasized (only matter linearly) in our loss function. The parameter δ is chosen to be:

$$\delta = 0.25\sqrt{\mathrm{TSS}_D} = 0.25\sqrt{\frac{1}{2N_pT_D}\sum_{i=0}^{(N_p-1)}\sum_{t\in\mathbb{T}_D}\sum_{\alpha}^{\{x,y\}}(\ddot{w}\circledast_t\vec{\rho}_i)_\alpha^2}, \quad \text{[S10]}$$

where TSS is total sum of squares of the experimental acceleration in the loss function. D refers to either train or test data set. To quantify the quality of the model's fit, we define R^2 as:

$$R^2 = 1 - \frac{\mathrm{RSS}_{\mathrm{test}}}{\mathrm{TSS}_{\mathrm{test}}}, \quad \text{[S11]}$$

where RSS is residual sum of squares:

$$\mathrm{RSS}_D = \frac{1}{2N_pT_D}\sum_{i=0}^{(N_p-1)}\sum_{t\in\mathbb{T}_D}\sum_{\alpha}^{\{x,y\}}(w\circledast_t\vec{f}_i^{\mathrm{env}} + \sum_j w\circledast_t\vec{f}_{ij} + \gamma_i\dot{w}\circledast_t\vec{\rho}_i - \ddot{w}\circledast_t\vec{\rho}_i)_\alpha^2. \quad \text{[S12]}$$

We note that for $R^2 > 0.99$, the average percentage error should be $\sqrt{1-R^2} < 10\%$. Therefore, we set an arbitrary threshold, $\delta = 0.25$, which indicates that data with an error that is 2.5 times the average error should be considered an outlier in the Huber loss. Finally, the data is split into 10 temporal sections, and 10 models are trained by 10-fold cross-validation. Such splitting ensures that models inferred in one section work in the others, so that there are no significant drifts in the experiments. Note that because of the convolution (Eq. S5), for a data with time length T, t can only be defined on $\tau/2 \le t < T - \tau/2$. For the l-th model, $\mathbb{T}_{\mathrm{test}} = \left\{t|\tau/2 + \frac{l-1}{10}(T-\tau) \le t < \tau/2 + \frac{l}{10}(T-\tau)\right\}$, and $\mathbb{T}_{\mathrm{train}} = \{t|\tau/2 \le t < T - \tau/2 \text{ and } t \notin \mathbb{T}_{\mathrm{test}}\}$. The average (test) R^2 of the 10 models for the 10-fold validation is reported in Table 1. The error bars shown in Figs. 3, 4 in the *Main text* are calculated from the standard deviation of the 10 models' prediction. We note that this estimation of the error bars only includes the variance of the model, plus the variation caused by the temporal plasma environment fluctuation in the experiments, while the bias of the model is excluded.

The time it takes to fully train our model scales as N_p^2 and typically takes 2-3 hours on desktop computer with an Intel 14900 processor. At the cost of accuracy and more complex book-keeping (see **Data processing** below), one can simply truncate the interaction force between two particles at large separations, so that the training time scales as N_p. However, this is not the most challenging part when extending our model to a large number of particles. The model's ability to infer forces requires a particle-level descriptor, s_i, meaning mis-identification or mis-tracking of particles can have a detrimental effect on the model's performance.

Data processing. To train the model, the data $x_{i,t}$, $y_{i,t}$, $z_{i,t}$, and $s_{i,t}$ needs to be organized into a form that can be efficiently iterated over to save computational time. Note that s_i is a time-averaged descriptor of particle i and is independent of t. However, for consistency in the input, we constructed the array of $s_{i,t} \equiv s_i$ at all times. We need three tensors that can be used to calculated the convolution of the data with w, $\dot{w}$, and $\ddot{w}$. Thus, each term in the loss function was associated with a separate tensor of data to compute the convolution. The data is first processed into three tensors X^0, X^1, and Y. Y is the target, which is a 3D tensor with shape of $N_p \times (T-\tau) \times 2$. $Y_{i,t,\alpha} = \ddot{w}\circledast_t\alpha_i$ where α is either x or y. Similarly, $X^1_{i,t,\alpha} = \dot{w}\circledast_t\alpha_i$. X^0 is a 5D tensor, with a shape of $N_p \times (T-\tau) \times (\tau-1) \times N_p \times 4$. To explain the meaning of $X^0_{i,t,t',k,\alpha}$, we first define an index function on $0 \le i < N_p$, $0 \le k < N_p$:

$$n(i,k) = \begin{cases} i, & \text{if } k = 0, \\ k-1, & \text{if } 0 < k \le i, \\ k, & \text{if } i < k < N_p. \end{cases} \quad \text{[S13]}$$

Then $X^0_{i,t,t',k,\alpha} = \alpha_{j,t+t'+1}$, where α can be x, y, z, or s, and $j = n(i,k)$. Here, note that when calculating $w\circledast_t f_{ij}$, only the input from time $t - \tau/2 + 1$ to $t + \tau/2 - 1$ is needed, with a total length of $\tau - 1$, because $w_{\pm\tau/2} = 0$. Finally, the first two dimensions of all three tensors are flattened, and the last two dimensions of X^0 are flattened, making X^0, X^1 and Y 3D, 2D, and 2D tensors, respectively.

Dusty plasma simulations. In order to test the accuracy of the ML methods, and the inference of the mass and charge of particles, we simulated our dusty plasma system using a custom molecular dynamics code. The simulations are similar to those used in previous studies (3–5). The simulations consisted of 15 spherical particles whose diameters were chosen from a Gaussian distribution with a mean of $d_0 = 10$ μm and a standard deviation of 1 μm. In the horizontal, xy-plane, the particles were confined by a harmonic potential with a small degree of asymmetry to match the experiments. They also experienced a vortical force to induce rotation of the system, leading to the following environmental reduced force:

$$f_{i,x}^{\mathrm{env}} = (1+\beta)\chi_h q_i x_i/m_i + \Omega^2 y_i - \gamma\dot{x}_i, \quad \text{[S14]}$$

$$f_{i,y}^{\mathrm{env}} = (1-\beta)\chi_h q_i y_i/m_i - \Omega^2 x_i - \gamma\dot{y}_i. \quad \text{[S15]}$$

The degree of asymmetry of the potential was determined by the dimensionless number β, χ_h is the electric field gradient, q_i and m_i are the charge and mass of particle i, Ω is the strength of the background vorticity from ion drag, and x_i and y_i are the horizontal coordinates of particle i. These parameters are all necessary to describe the general environmental confinement of dust particles, as discussed in detail in Ref. (5). Dotted variables indicate differentiation with respect to time and the Epstein drag force is determined by γ. The mass of each particle was computed as $m_i = \rho_p \pi d_i^3/6$, where $\rho_p = 1{,}510$ kg·m^{-3}, and d_i is the diameter of particle i.

In the vertical direction, the particles experienced forces due to a linearly-varying electric field, and gravity. The reduced force was determined by the following equation:

$$f_{i,z}^{\text{env}} = \min(E_0 + \chi_z z_i, 0) q_i/m_i - g - \gamma \dot{z}_i + \eta w(t). \quad \text{[S16]}$$

Here E_0 is a constant vertical electric field, χ_z is the electric field gradient, z_i is the vertical position of the particle, and g = 9.81 m·s^{-2} is the acceleration due to gravity. The **min** function guarantees that the electric force will never change sign, and thus the edge of the plasma sheath occurs at $z_{\text{edge}} = -E_0/\chi_z$, a small distance above $z = 0$. The last term provides a small amount of stochastic noise in the z direction. This noise drives oscillations in z since the particles behave as stochastic harmonic oscillators with a well-defined resonance frequency. The function $w(t)$ represents a Wiener process with zero mean and unit standard deviation, and η is the strength of the noise. Since we are not inferring forces in z, this does not affect the inference procedure, and is based on previous experiments in our lab illustrating z oscillations originating from Brownian motion (5) and spontaneous oscillations due to delayed charging at low pressures (6). We also allow the charge on the particle to vary linearly within the sheath, increasing in magnitude as z_i decreases. This was done by treating each particle as a spherical capacitor, and parameterizing the charge in the following way:

$$q_i = \min(2\pi\epsilon_0 d_i V(1 - z_i/l_q), -8 \times 10^{-16}), \quad \text{[S17]}$$

where the units of charge are in Coulombs. This guarantees that the magnitude of the (negative) charge on the particle will never be smaller that 5,000e, and the magnitude of charge increases deeper into the sheath (smaller z_i). Here l_q is a length scale that determines the strength of charge variation in the sheath. The voltage V is a constant that determines the charge on a particle at $z = 0$.

The parameters described here, such as electric field, are difficult to relate to experimental measurements. Thus, we fixed these parameters by relating them to the typical frequencies of small oscillations of the particles around their equilibrium positions. Experimentally, these can be measured from the 3D tracking data (7), and are given by ω_h in the horizontal direction, and ω_z in the vertical direction. Linearizing the force around $z = 0$, so that $f_z^{\text{env}} = 0$, $\omega_h^2 = -df_x^{\text{env}}/dx$, and $\omega_z^2 = -df_z^{\text{env}}/dz$, we arrive at the following relationships:

$$\chi_h = -\frac{\rho_p d_0^2 \omega_h^2}{12 V \epsilon_0}, \quad \text{[S18]}$$

$$E_0 = \frac{\rho_p d_0^2 g}{12 V \epsilon_0}, \quad \text{[S19]}$$

$$\chi_z = \frac{\rho_p d_0^2 (g - l_q \omega_v^2)}{12 l_q V \epsilon_0}. \quad \text{[S20]}$$

This way a particle with diameter d_0 would have its equilibrium position at $z = 0$, and frequencies of small oscillations exactly equal to ω_h and ω_z. However, since particle sizes are drawn from a Gaussian distribution centered at d_0, the frequencies vary as well.

In addition to the environmental forces, the particles experienced a pairwise, non-reciprocal repulsive force. This force stems from basic Coulomb repulsion, but also from the wake of ions streaming past each particle. As done in Ref. (8), we parameterized this ion wake by an effective positive cloud of charge with magnitude $\tilde{q} q_i$ at a distance h beneath each particle. The force between particles was derived from the following potential:

$$\phi(\vec{r}) = \frac{q_i q_j}{4\pi\epsilon_0 \lambda_D} \left[\frac{e^{-r/\lambda_D}}{r/\lambda_D} - \tilde{q} \frac{e^{-r_w/\lambda_D}}{r_w/\lambda_D} \left(1 + b \frac{e^{-r_w/\lambda_D}}{r_w/\lambda_D} \right)^{-1} \right]. \quad \text{[S21]}$$

Here, $\phi(\vec{r})$ is the potential of the ith particle in the field of the jth particle and its wake, and $f_{ij} = -\vec{\nabla}_i \phi$. The position vector between the particles is $\vec{r}$, $r_w = |\vec{r} - h\hat{z}|$ is the distance from particle i to the wake of particle j, $\hat{z}$ is the unit vector in the z direction, λ_D is the Debye screening length in the plasma, and b is dimensionless cutoff used to truncate the divergence of the wake interaction since the wake is not a point charge, but more of a cloud. With these environmental and interaction forces, the Newton's 2nd law was integrated forward in time using the 2nd-order velocity Verlet method.

Without energy input, Epstein drag would drain the energy from the system and the particles would assume equilibrium positions. However, there are three mechanisms that drive kinetic and potential energy into the particles' motion. The first is the vortical force from ion drag, which is non-conservative. The second is the small amount of stochastic noise in the z-direction. The third is the non-reciprocal interaction force (also non-conservative) (9). For a given simulation, we chose parameters that produced particle motion most visually similar to the experiments, or parameters that could be measured

directly from experiments (like oscillation frequency). Movie S6 shows that the resulting motion of the particles indeed looks strikingly similar to the experiments, and can be easily analyzed by our ML model.

Prior to training the model, Gaussian-distributed measurement error with standard deviation 0.005 mm was added to each particle position to simulate experimental particle tracking error. In our simulation, we used $\lambda_D = 0.8$ mm (Eq. S21) for all particles. Figure S2*A* shows that at the same z-position, there is only a weak dependence of the fitted effective screening length on different particle sizes since λ only varies from 0.48 - 0.52 mm for different particle pairs. This indicates that the particle-dependent effective screening length λ in experiments is real (Fig. 4*A*), rather than an artifact of the ML model. Moreover, the predicted interaction agrees with the exact interaction with less than 10% error (Fig. S2*A*). Even though the fit is very good, as discussed in the main text, the presence of a virtual ion wake can systematically reduce the fitted values of the screening length ($\lambda = 0.52$ mm from the fit, and $\lambda_D = 0.80$ mm in Eq. S21).

Figure S2*B* shows the inferred masses from the damping term, assuming Epstein drag (Eq. 2 in the *Main Text*), and the mass inferred from the fitting procedure, versus the actual masses of particles used in the simulation. The agreement is remarkable and demonstrates that our model can accurately infer each term in the equation of motion. Figure S2*C* shows the inferred charge on each particle versus the inferred mass. The fitted slope of $p = 0.31$ is close to the expected value from the simulation, $p = 1/3$, and reflects the fact that particles at the same vertical position will have the same floating potential, independent of their mass (Eq. S17). However, fitting to Eq. 4 in the *Main Text* results in a deviation of the prefactor $A = 89$ mm^3·s^{-2} from the actual $q_1 q_2/4\pi\epsilon_0 m_1 = 103$ mm^3·s^{-2}. This deviation of A can cause the inferred q to be systematically lower than the actual q by 5-10%.

Taken together, Fig. S2 suggests that the inference of interaction forces in simulated data is excellent, ion wake-mediated interactions can significantly reduce the effective screening length, and the inference of particle charge is very good (5-10% error).

Data availability

All 5 experimental 3D trajectories and simulated trajectories, plus the code of our machine-learning algorithm, is available on github: https://github.com/wyu54/many-body-force-infer.

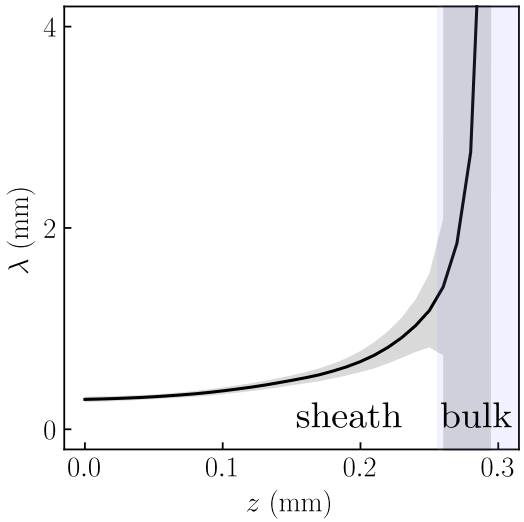


Fig. S1. The fitted screening length λ versus z for the 20 interactions of particles 1 and 2. The gray shaded region indicates uncertainty. Note that our model's accuracy can only infer interaction when $\rho < 1.2$ mm, and the parameter λ is introduced in the term of $e^{-\rho/\lambda}$ in Eq. 4 in the *Main Text*. Thus when $\lambda > 1.2$ mm, the fitting cannot give accurate predictions of λ. The purple shaded region indicates bulk plasma, in which λ is predicted to be larger than 1.2 mm.

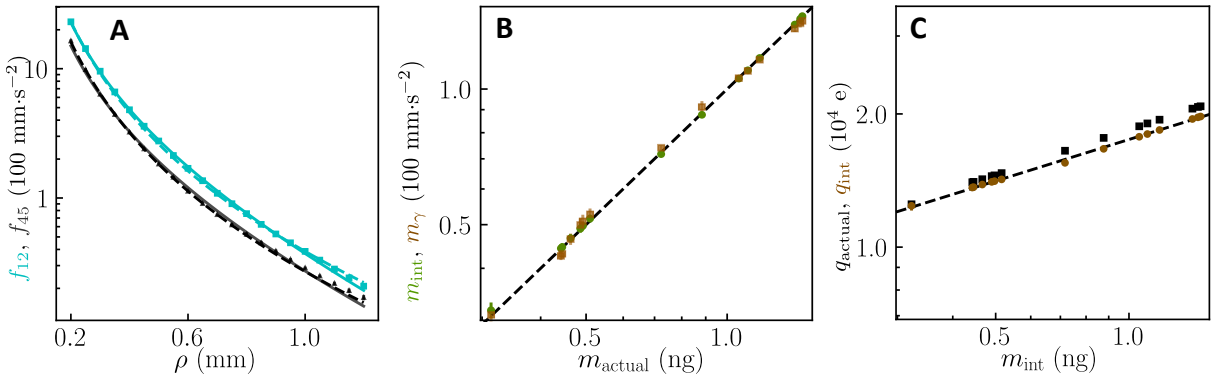


Fig. S2. Results from the inference of mass and charge in dusty plasma simulations. The parameter values used for this simulation were N = 15, d_0 = 10.0 μm, λ = 0.8 mm, Ω = 4.25 Hz, V = -5 V, ω_h = 21.2 Hz, ω_v = 157 Hz, β = 0.1, l_q = 1 mm, η = 5 m/s^2, $\tilde{q} = 0.5$, and $b = 0.2$. (**A**) The interaction between two small particles indexed 1 and 2 (blue squares, the actual masses used in the simulation $m_1 = 0.44$ ng and $m_2 = 0.46$ ng), and two big particles indexed 4 and 5 (gray triangles, actual masses $m_4 = 1.43$ ng and $m_5 = 1.45$ ng). The solid lines with corresponding colors are fits to Eq. 4 in the *Main Text* with parameters A = 97 mm^3·s^{-2}, $\lambda = 0.48$ mm (f_{12}), and A = 64 mm^3·s^{-2}, $\lambda = 0.52$ mm (f_{45}). The dashed lines are the actual interaction reduced force used in the simulation. (**B**) The mass of all particles inferred from the drag coefficient (m_γ, green circles) and the mass inferred from the particle interactions ($m_{\rm int}$, brown squares), versus their actual mass used in the simulation. The dashed line represents $m_{\rm int} = m_\gamma = m_{\rm actual}$. (**C**) Particle charges inferred from their interactions ($q_{\rm int}$, brown circles), and their actual charges used in the simulation ($q_{\rm actual}$, black squares), versus $m_{\rm int}$. The dashed line is the optimal power-law fit to the inferred data with power $p = 0.31$.

Movie S1. 3D motion of $N_p = 9$ particles, as labeled in Table 1 in the main text.

Movie S2. 3D motion of $N_p = 10$ particles, as labeled in Table 1 in the main text.

Movie S3. 3D motion of $N_p = 13$ particles, as labeled in Table 1 in the main text.

Movie S4. 3D motion of $N_p = 15$ particles, as labeled in Table 1 in the main text.

Movie S5. 3D motion of $N_p = 18$ particles, as labeled in Table 1 in the main text.

Movie S6. 3D motion of $N_p = 6$ particles showing a rotating crystal state.

Movie S7. 3D motion of $N_p = 9$ particles showing a rotating crystal state.

Movie S8. 3D motion from our simulation of a dusty plasma system with non-reciprocal forces, as described in the SI

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